

HoloBurner:

Mixed Reality Equipment for Learning Flame Color Reaction by using Aerial Imaging Display

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ABSTRACT

The HoloBurner system is a mixed reality instrument of chemistry experiments for learning flame color reaction. It can provide users with the sense and enjoyment of performing experiments in a laboratory. Flame color reaction is a phenomenon which can identify an element by its unique color, with which students can learn that substances are comprised with various different elements. The HoloBurner system provide users a real feeling of operating a burner, and realistic image of flame with an aerial imaging display. Students can experience flame color reaction using the HoloBurner safely in natural way same as with real instruments.

CCS CONCEPTS

• **Applied computing** → Education; Interactive learning environments; • **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Mixed / augmented reality.

KEYWORDS

Physical Mixed Reality, chemistry education, burner, flame color reaction

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1 INTRODUCTION

Today's the digital educational environment offers a variety of useful educational tools and can provide learners with appropriate learning experiences regardless of time and place. On the other hand, there is also an emphasis on having opportunities for physical experiences, but safety-critical experiments tend to be avoided. In addition, with the spread of coronavirus infections, opportunities for physical experiences are being further lost.

We are working on a physical Mixed Reality project that uses Mixed Reality technology to integrate the virtual into the real world and extend the experience through user interfaces embedded in

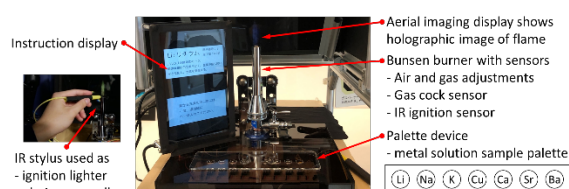


Figure 1: HoloBurner system.

physical space. In this project, we are developing physical Mixed Reality displays that allow users to interact naturally with virtual, digital, or conceptual things in a physical context. The digital and virtual objects include captured, digitized, and fully synthesized objects, physical and chemical phenomena, mathematical formulas, computational algorithms, and simulated artificial life [Ohshima, 2018]. The users are able to physically touch, feel, and share the digital entities in the real world. This will allow users to extend their experience beyond physical constraints and limitations. The main goal of this project is to create new educational tools in various fields such as chemistry, biology, physics, mathematics, programming and so on.

The HoloBurner system is a physical mixed reality device for chemical experiments to learn how to safely operate a gas burner and flame color reactions. The flame color reaction is a phenomenon that allows us to identify elements by their unique colors. With a simple experiment, students can learn that matter is made up of various elements. The flame color reaction also tells us why fireworks and nebulae come in different colors. Through the virtual use of a real burner, students can learn about the properties of flammable gases, safety awareness and attitude, and the fun of chemical experiments.

2 HOLOBURNER SYSTEM

HoloBurner System consists of an aerial imaging display, a burner device, a palette device and a stylus as shown in Figure 1. An instruction display monitor shows users how to use the instruments and explains about flame color reaction.

The burner device consists of a real Bunsen burner and a sensor attached to it. It provides a realistic operating feel. Users can learn the correct way to operate the burner from ignition to extinguishing. Rotary sensors detect the rotation of the adjustment screws. A photo sensor is inside the tube, and the burner is ignited by an infrared light stylus.

AIRR [Yamamoto, 2014], a retro-reflective imaging technique, is used to display the aerial image. Users can see floating image of flame at the tip of the burner as seen in Figure 2. The viewing angle

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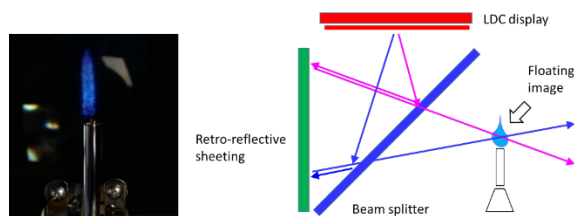


Figure 2: Aerial imaging display.

Table 1: The colors of the flames.

Element	Color	Primary wavelengths[nm]
Li	Red	610.4, 670.8
Na	Yellow	589.0, 589.6
K	Purple	404.4, 766.5, 769.9
Cu	Green	(537)
Ca	Orange	422.7, (554), (622)
Sr	Red	460.7, (605)
Ba	Yellow-Green	455.1, 493.4, 553.5

from which the user can see the flame is within about 80 degrees horizontally, centered on the flame image. This limitation is mainly due to the horizontal size of the retroreflective sheet and the display device, and there is room for improvement. We also developed a biology learning mixed reality device by using AIRR [Ohshima, 2018]. In addition to AIRR, an approach using micro-mirror arrays [Maekawa et al., 2006] would be possible in principle.

The palette device is a user interface for selecting a sample solution of metal salts, and there are seven different sample solutions. There are eight photo sensors, each of which corresponds to a sample. And one of them corresponds to hydrochloric acid solution to clean up a platinum needle. The infrared light stylus acts as a platinum needle to pick up the sample solution. An infrared LED is placed inside, and a photo sensor placed in the palette detects the light and determines which element is selected.

The colors of the flames are approximated to be as close to sensible as possible for each element as shown in Table 1. Flame is simulated according to status of adjust screws and expressed by using a particle system method.

3 EXPERIENCE

There are three steps to operating the instrument.

- Step 1. Ignite the burner
- Step 2. Flame color reaction experiment
- Step 3. Extinguish the flame of the burner

The guide monitor next to the burner will show the users the correct way to operate the burner and warn them if they do it incorrectly. It would be an even more natural interface if the operation of the burner and warnings of mistakes were displayed as aerial images near the burner. Figure 3 shows steps of ignite and turn of the burner flame.

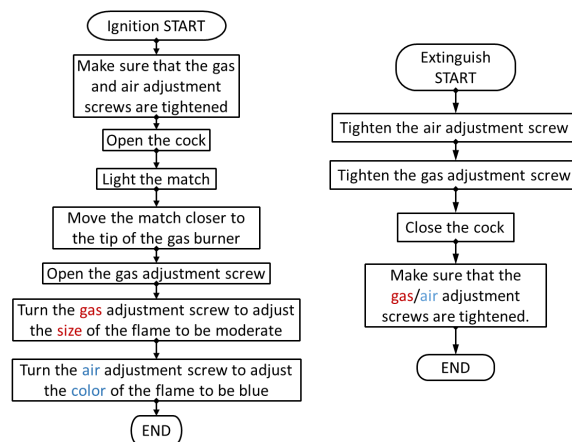


Figure 3: Ignite and extinguish the burner.

4 CONCLUSION

We proposed a new hands-on learning material for chemistry education. We created the HoloBurner system in order to experiment with flame color reaction. The HoloBurner system provides an immersive experience as if you are actually operating the burner, and realistic images of the flame through an aerial display. The mixed reality of the physical burner and the holographic flame has potential applications in science education for various chemical experiments.

With the current system, users can perform virtual experiments on their local sites off-line by using the mixed reality instrument. In the next step, the physical instrument will be connected with remote sites of other students and/or teachers on-line. Our mixed reality physical instrument is expected to enhance communication and education via both of on-line and off-line.

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